

High-Performance Speed Measurement by Suppression of Systematic Resolver and Encoder Errors

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Abstract—The subject of this paper is a method which suppresses systematic errors of resolvers and optical encoders with sinusoidal line signals. The proposed method does not require any additional hardware and the computational efforts are minimal. Since this method does not cause any time delay, the dynamic of the speed control is not affected. By means of this new scheme, dynamic and smooth running characteristics of drive systems are improved considerably.

Index Terms—Adaptive control, motion control, motor drives, position measurement, velocity control, velocity measurement.

I. INTRODUCTION

FOR PRECISE motion control, the quality of the speed feedback is one of the most relevant bottlenecks to achieving high performance. This becomes obvious when the existing limitations are considered that hinder higher dynamics and better smooth running characteristics of the drive system.

Imperfections of the mechanical system, such as elasticity and backlash, reduce the attainable stiffness of the speed control. In order to avoid these mechanical imperfections and to gain higher drive performance, the use of direct drives [1] is becoming more and more common. For direct-drive applications the inertia is usually high relative to nominal torque, but the high gains, which theoretically result for the speed and position controllers according to standard design rules, often have to be reduced in practice due to the insufficient performance of the measurement signals.

If direct drives cannot be used in the application, the stiffness of the speed control loop can be improved by control schemes adapted to the imperfections of the mechanical system [2]. The respective control structures use observers to gain information about the nonmeasurable states. The observers' performances, again, rely mainly on the speed signal quality.

Other limitations of the dynamics and smooth running characteristics are caused by delay times in the speed control loop. The delay times cover the dynamic behavior of the current control, a jitter filter for speed measurement and the sample time of the digital control algorithms. The delay time of the current control can be decreased only with difficulties, since it depends mainly on the switching frequency of the inverter

and the switching frequency cannot be further increased, due to switching losses. Modern powerful signal processing already allows sampling times of the speed control algorithms that are lower than the time delay of the current control. With regard to this, the delay time, caused by the jitter filter of the speed measurement, offers the main potentiality for the reduction of time delays.

To this end, the improvement of the speed measurement system becomes the key aspect for enhanced dynamics and smooth running characteristics. The resulting challenge is to design speed measurement systems offering high resolution and accuracy as well as short delay and dead times.

Position and speed measurement in motion control servo drive systems is most frequently performed by using resolvers. For higher performance, optical encoders with sinusoidal line signals are preferred. After demodulation of the resolver signals, position information is determined from two sinusoidal, 90° phase-shifted line signals for both. For example, the position signal is calculated by counting the number of periods and interpolating within one period. The speed is determined by deriving the position.

Unfortunately, the line signals contain various errors, which result in position and speed inaccuracies, if they are disregarded at the interpolation. Depending on the frequency of the speed errors, either vibrations of the mechanical system are stimulated, or the speed control follows these errors with a bad impact on the smooth running characteristics.

Many approaches have been proposed to improve speed measurement performance. Gees [3] uses an additional acceleration sensor. This allows noise suppression because the speed is determined by integrating the acceleration signal. The disadvantages are the high expenses for sensor and wiring. Burke [4] and Emura [5] utilize phase-locked loop (PLL) techniques that either incur additional hardware expenses or require high computing power. Besides this, the performance at low speed is not improved. The same is true for the frequency-shift method [6] proposed by the authors. Kirchberger [7], however, utilizes an oversampling method. By this method high frequency errors are suppressed by averaging. Similar to the PLL technique, this method requires high hardware expenses and performance at low speeds is not improved. The software-based compensation of encoder errors proposed by Hoescheler [8] is closer to the practical requirements of drive systems. Remaining disadvantages are the need for an adjustment of the compensation on every resolver or encoder and the missing compensation of long-term drifts by self-tuning methods.

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In this paper the gain-phase-offset-correction method (GPOC) is introduced, which requires no additional hardware and only low computing power. The most important errors are suppressed at any speed by a self-tuning method.

II. SYSTEMATIC ERRORS OF RESOLVERS AND ENCODERS

The following considerations are simplified by combining the two line signals s_a and s_b of resolvers or encoders to one complex signal $\underline{s}(\varepsilon) = s_a + js_b$. Assuming that:

- random noise (caused, e.g., by quantization errors) can be neglected and
- the line signals are periodical functions of the position angle ε

this complex line signal can be represented by its Fourier series

$$\underline{s}(\varepsilon) = \sum_{k=-\infty}^{\infty} \underline{c}_k e^{jk\varepsilon}. \quad (1)$$

Since the zero mark depends only on the mechanical fitting of the encoder it is always possible to determine an encoder offset, so that the coefficient $\underline{c}_1$ becomes real, $\underline{c}_1 = c_1$. Additionally, for most encoders offering only small errors, assumption $c_1 \gg |\underline{c}_{k,k \neq 1}|$ can be made, whereas for an ideal encoder even $\underline{c}_{k,k \neq 1} = 0$ is true. In (1), the coefficient $\underline{c}_0$ describes the offset errors of the line signals, $\text{Re}\{\underline{c}_{-1}\}$ an amplitude difference of both line signals and $\text{Im}\{\underline{c}_{-1}\}$ a phase error of the line signals.

If the encoder is not ideal ($\underline{c}_{k,k \neq 1} \neq 0$) and only the fundamental component c_1 is considered for interpolation, an error $\Delta\varepsilon$ for the measured position angle results, which is given by

$$\begin{aligned} \Delta\varepsilon &= \arg(c_1 e^{j\varepsilon} + \underline{c}_k e^{jk\varepsilon}) - \varepsilon \\ \Delta\varepsilon &= \arg\left(c_1 + \underline{c}_k e^{j(k-1)\varepsilon}\right) \\ \Delta\varepsilon &\approx \frac{\text{Re}\{\underline{c}_k\}}{c_1} \sin((k-1)\varepsilon) + \frac{\text{Im}\{\underline{c}_k\}}{c_1} \cos((k-1)\varepsilon). \end{aligned} \quad (2)$$

Maximum limit values for offset errors of optical encoders are about 4% related to c_1 . The same is true for the amplitude differences. The harmonics $\underline{c}_{k,|k| \geq 2}$ however are usually less distinct, resulting typically to total harmonic distortions (THDs) less than 1%. Regarding this, the offset, amplitude, and phase errors are dominant and the harmonic distortion can be neglected.

Fig. 1 illustrates an example for the position errors caused by an erroneous interpolation. Their effects on the drive performance mainly depend on the number of encoder periods per turn, the absolute speed, and the sampling rate of the position measurement.

For resolvers, the number of periods per turn is small (typically $n = 1 \dots 3$), which leads to the following consequences.

- The absolute position error $\Delta\alpha = \Delta\varepsilon/n$ cannot be neglected and limits the positioning accuracy.
- Even at highest speeds, the Shannon Theorem is fulfilled concerning the position error harmonics.
- At low speeds, the bandwidth of the speed control is higher than the frequency of the speed errors. Thus, the speed and the position control follow the faulty measured speed.

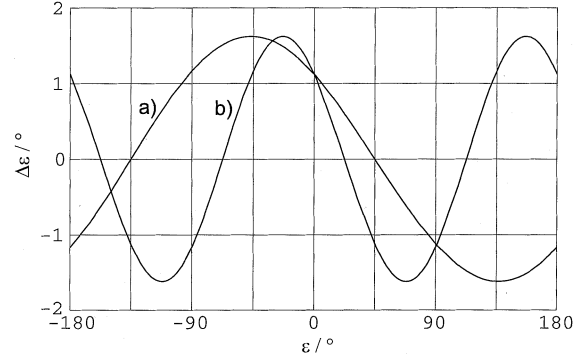


Fig. 1. Interpolation error $\Delta\varepsilon(c_1 = 1)$: (a) offset errors $\underline{c}_0 = 0.02(1 + j)$, (b) amplitude and phase errors $\underline{c}_{-1} = 0.02(1 + j)$.

For optical encoders the conditions are totally different, because the number of periods per turn is much higher (typically $n = 2^9 \dots 2^{13}$).

- The absolute position error $\Delta\alpha = \Delta\varepsilon/n$ is small and the positioning accuracy is nearly unaffected.
- Even at low and medium speeds, the position error includes higher frequencies than half of the sampling rate $1/T_S$ and the Shannon Theorem is broken. As a consequence the speed and position control can be disturbed at any frequency up to $1/(2T_S)$. Thus, a digital low-pass filter for speed measurement is only effective for a part of the error signals. The error components, appearing at low frequencies after sampling, cause oscillations in speed and position control.

III. SUPPRESSION OF SYSTEMATIC ERRORS

Taking these consequences into account, it becomes obvious, that system behavior can be considerably improved, if the dominant gain, phase, and offset errors are suppressed. Due to their small amplitudes, the harmonics of the line signals are neglected in the following.

Once the phase error φ_c , the amplitude correction factors v_a and v_b , as well as the offset errors $\underline{q}$ are determined, the corrected line signals $\underline{s}_c$ can be calculated by

$$\begin{aligned} \underline{s}_c &= \begin{pmatrix} 1 & 0 \\ -\tan \varphi_c & \frac{1}{\cos \varphi_c} \end{pmatrix} \cdot \begin{pmatrix} v_a & 0 \\ 0 & v_b \end{pmatrix} \cdot (\underline{s} + \underline{q}) \\ \underline{s}_c &\approx \begin{pmatrix} v_a & 0 \\ -v_a \varphi_c & v_b \end{pmatrix} \cdot (\underline{s} + \underline{q}). \end{aligned} \quad (3)$$

It has to be considered that the phase error is ambiguous. Therefore, it cannot be decided, which line signal shows the respective error.

Now, a new scheme is derived, which determines these correction terms. To gain a high acceptance in the field of commercial drives it shows the following features.

- No additional hardware efforts are required.
- Standard analog digital converters for drive applications without oversampling facilities are used.
- Additional computation power is low and the scheme has been implemented on a low-cost 16-bit microcontroller.
- The method is suited for resolvers and optical encoders.

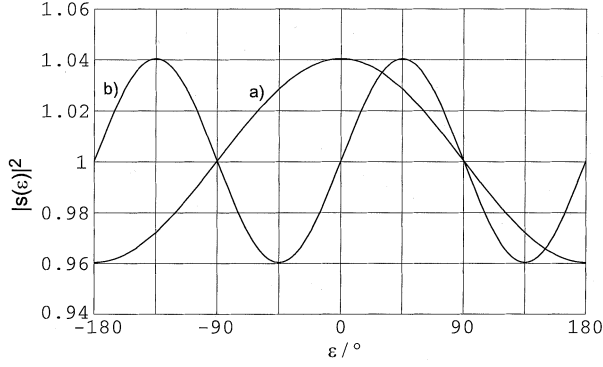


Fig. 2. Squared magnitude of $\underline{s}(c_1 = 1)$: (a) $\underline{c}_0 = 0.02(1+j)$ and (b) $\underline{c}_{-1} = 0.02(1+j)$.

- The method is applicable in a wide speed range.
- The method adapts online to time-varying encoder errors.
- No complicated commissioning of the scheme is necessary.

The proposed method analyzes the squared magnitude of the line signals and utilizes the orthogonality of trigonometric functions.

For an ideal encoder, the squared magnitude given by

$$|\underline{s}(\varepsilon)|^2 = \sum_{k=-\infty}^{\infty} \underline{c}_k e^{jk\varepsilon} \cdot \sum_{k=-\infty}^{\infty} \underline{c}_k^* e^{-jk\varepsilon} \quad (4)$$

is constant for any position angle.

With the assumptions discussed above, namely, c_1 is real and $c_1 \gg |\underline{c}_{k,k \neq 1}|$, this product can be simplified as follows:

$$\begin{aligned} |\underline{s}(\varepsilon)|^2 &\approx c_1^2 + \sum_{\substack{k=-\infty \\ k \neq 1}}^{\infty} c_1 \left(\underline{c}_k^* e^{-j(k-1)\varepsilon} + \underline{c}_k e^{j(k-1)\varepsilon} \right) \\ |\underline{s}(\varepsilon)|^2 &\approx c_1^2 + \sum_{\substack{k=-\infty \\ k \neq 1}}^{\infty} 2c_1 \operatorname{Re}\{\underline{c}_k\} \cos(k-1)\varepsilon \\ &\quad - \sum_{\substack{k=-\infty \\ k \neq 1}}^{\infty} 2c_1 \operatorname{Im}\{\underline{c}_k\} \sin(k-1)\varepsilon. \end{aligned} \quad (5)$$

This result, illustrated in Fig. 2, is interesting under various aspects.

- Each component $\underline{c}_{k,k \neq 1}$ causes amplitude variations with frequencies at the $|k-1|$ harmonics of the fundamental.
- If an amplitude variation of the n th harmonic appears, it can be caused either by the $(n+1)$ or the $(1-n)$ harmonic of the line signals.
- If one of both possible line signal harmonics can be neglected, the coefficient $\underline{c}_k$ can be determined unambiguously from the time signal of the squared magnitude.

To determine these coefficients, the cross-correlation method is well suited, which allows a good suppression of stochastic error signals and is able to cancel completely the influences of undesired harmonics.

For cross correlation the integral

$$\tilde{\underline{c}}_k = c_1 \underline{c}_k = \int_0^{n2\pi} |\underline{s}(\varepsilon)|^2 e^{j(k-1)\varepsilon} d\varepsilon \quad (6)$$

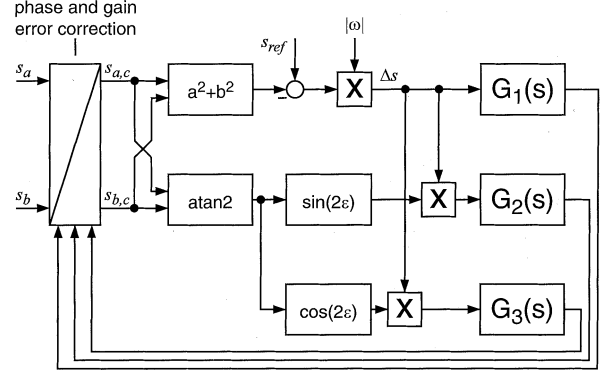


Fig. 3. New scheme for suppressing gain and phase errors of sinusoidal line signals.

has to be calculated. The integration variable, position angle $d\varepsilon$, can be replaced by the product of the actual speed of the drive ω and the time t , $d\varepsilon = \omega dt$, which simplifies a time discrete calculation.

Besides this, calculation efforts are further reduced, when integration is not performed exactly over one period and the respective feedforward-control-oriented scheme, given by (6), is replaced by a feedback scheme using I-controllers. In this case the time constants of the I-controllers have to be sufficient large, because the correction values can not be exactly determined at any time and their mean values have to be calculated over a sufficient number of periods. However, a large time constant is not critical, because the controller has to react only to slow drift effects of the encoder signals.

Following this approach the control scheme depicted in Fig. 3. results, which determines the correction terms of phase and amplitude errors.

The squared magnitude is calculated from the corrected line signals, see (3), and the actual interpolation angle ε is calculated, which can be utilized for position measurement.

The squared magnitude is compared to the respective reference and an error signal is fed back via the amplitude controller to adjust the amplitude correction factors. The amplitude difference is weighted by the terms $\sin 2\varepsilon$ and $\cos 2\varepsilon$, and the respective signals are given to the controllers for phase and amplitude error. The control scheme of Fig. 4 shows an offset correction using the same scheme. Cross correlation of the squared magnitude is performed simply with the line signals. Note that all five controllers needed for the error correction are I-controllers which results in minimal computational efforts.

Of course, applications exist where the assumptions are not valid and the proposed online adaptation method is not practical. Examples are fast point-to-point positioning with small distances, where less than one period of the resolver's line signals can be evaluated. For encoders, operation is critical at constant speeds, which result in frequencies of the line signals close to $f = 1/(2T_S)$ or respective harmonics. In these cases only the online adaptation has to be suspended. The correction of the errors itself can be performed with the last determined values continuously. However, an offline tuning of the errors is still possible in all applications, where the speed of the drive can be chosen freely during commissioning.

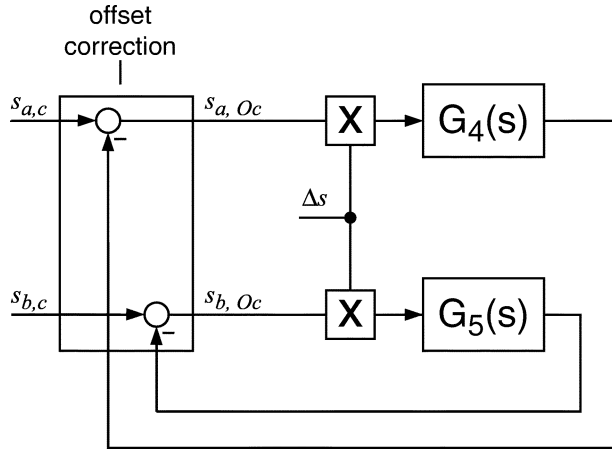


Fig. 4. Scheme for suppressing offsets of sinusoidal line signals.

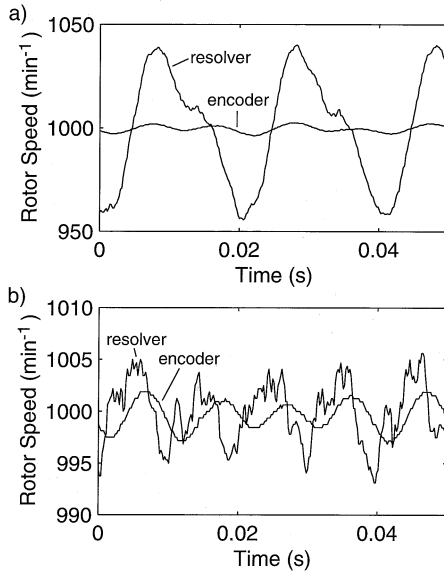


Fig. 5. Results obtained with resolver, (a) without and (b) with correction.

IV. PRACTICAL RESULTS

The proposed method has been integrated into a servo controller drive, the CDD 3000 of Lust Antriebstechnik, Lahnau, Germany. All calculations are performed by a simple 16-bit microcontroller in a few microseconds. No additional hardware components are needed.

A. Results Obtained With Resolvers

The effects of the proposed new method on servo drives equipped with resolvers has been examined on a 0.47-kW synchronous motor drive with a rated speed of 3000 r/min. The motor and resolver both have six poles.

For the validation of the measurements this motor is also equipped with an additional optical encoder (1024 periods per turn). Because of its higher performance compared to the resolver under test, this measuring system provides nearly the actual speed of the drive. Therefore, this speed, calculated from the encoder signals, can be regarded as a reference.

One result, obtained at a constant motor speed of 1000 r/min, is illustrated in Fig. 5. The corresponding and demodulated line

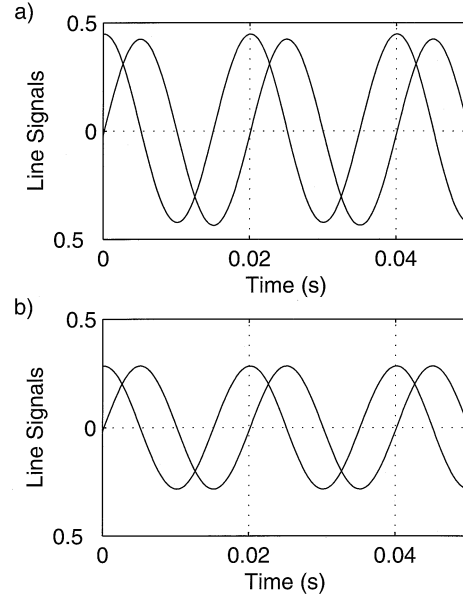


Fig. 6. (a) Original line signals and (b) corrected line signals of resolver system.

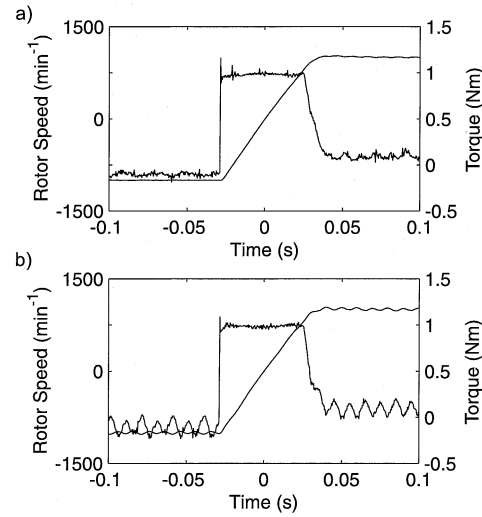


Fig. 7. Speed reversal with resolver, (a) with correction and (b) without correction.

signals are shown in Fig. 6. In spite of the small errors of the line signals, the speed error is enormous. According to (2), it can be derived, that the offset causes the dominant error in this case, because the first harmonic makes the main contribution to the speed error. With enabled error correction and online adaptation the behavior can be improved significantly and the speed error is decreased by about a factor of 10. The offset error determined by the correction scheme is about 2.4% and -1.2% for both line signals. The differences of the amplitudes are about 1.2% and the phase error is about 0.25° .

The good performance of the correction scheme at a speed reversal is shown in Fig. 7. A slow zero crossing of the speed has been achieved by limiting the torque to 1 N·m. Even the zero crossing of the speed with corresponding low frequencies of the line signals of the resolver does not affect the correction

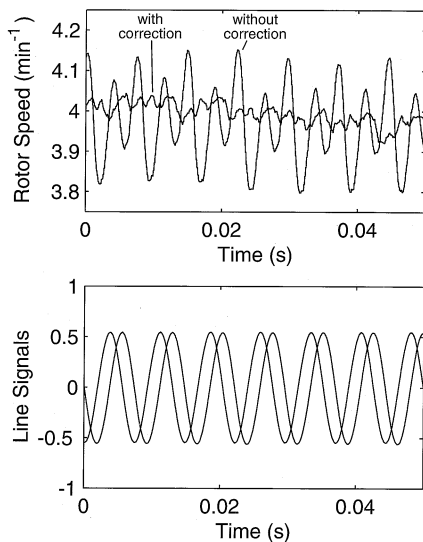


Fig. 8. Measured speed and line signals of encoder system.

scheme as well as the self-tuning. The smooth running has been improved over the whole speed range.

B. Results Obtained With Encoders

Further experiments have been performed on a 3.6-kW induction motor drive, with a rated speed of 1500 r/min. The motor is equipped with an optical encoder system (2048 periods per turn). Since no measuring system with a significant higher performance has been available, no speed signal can be given as a reference. On the other hand, variations of the instantaneous speed with the first and the second harmonics of the line signals frequency can be regarded as measurement errors, because the inertia prevents that the mechanical system oscillates with these high frequencies.

Fig. 8 illustrates results at a constant speed of 4 r/min. Even if the variations of the speed determined from the uncorrected line signals are low, it shows frequencies corresponding to the line signals. With enabled error correction and online adaptation the behavior can be improved significantly and the speed error is decreased by about a factor of 5. The offset error determined by the correction scheme is about -0.9% and -3% for both line signals. The difference in amplitudes is about 0.4% and the phase error is about 3.7° .

V. CONCLUSION

In this paper the fundamental impact of the speed measurement system on the smooth running characteristics and the dynamic behavior of modern servo-controlled drives has been shown. For high-performance drives it is very important to suppress the line signals' offset errors and amplitude errors as well as phase errors. To this end, the new method GPOC has been presented. Special benefits of this new scheme are the minimal computational efforts and the online tuning ability.

The correct functioning and the outstanding performance of the GPOC method have been shown by measurements of the

servo controller drive CDD3000 of Lust Antriebstechnik. For resolvers as well as for encoders the smooth running characteristics can be enhanced enormously. The benefits of this scheme have already been proven in various industrial applications, especially in drives featuring high inertia, e.g., direct drives used for handling machines and lifts, and in several drives featuring low damped eigenfrequencies due to elastic couplings.

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